

Dog Breed Identification Using Transfer Learning

Project Description

This project focuses on identifying dog breeds from images using Transfer Learning and Deep Learning techniques. Instead of training a CNN from scratch, pre-trained models such as MobileNetV2 or ResNet50 are used to improve accuracy and reduce training time. The model is trained on labeled dog images and can classify different dog breeds accurately.

Objectives

- Build a deep learning model to classify dog breeds.
- Apply transfer learning for better accuracy and faster training.
- Perform data preprocessing and augmentation.
- Evaluate model performance using metrics.
- Deploy the trained model using a web interface.

Project Flow

- Data Collection – Download dataset and organize images by breed.
- Data Preprocessing – Resize, normalize, and augment images.
- Model Building – Load pre-trained model and add custom layers.
- Model Training – Compile and train with Adam optimizer and categorical crossentropy.
- Model Evaluation – Evaluate using accuracy, confusion matrix, and classification report.
- Model Deployment – Save model and build web app using Flask or Streamlit.

Project Structure

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Dog-Breed-Identification/  
■  
■■■ dataset/  
■ ■■■ train/  
■ ■■■ validation/  
■ ■■■ test/  
■  
■■■ notebooks/  
■ ■■■ EDA.ipynb  
■  
■■■ models/  
■ ■■■ dog_breed_model.h5
```

- - static/
 - ■■■■ uploaded_images/
 - - templates/
 - ■■■■ index.html
 - - app.py
 - train.py
 - requirements.txt
 - README.md